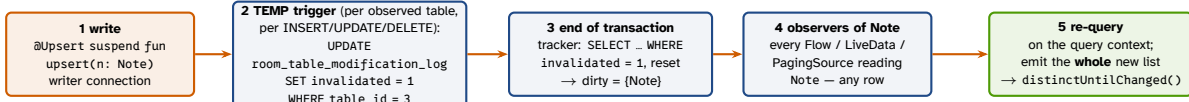


In one line: Room is a **compile-time** layer over SQLite: KSP reads `@Entity/@Dao/@Database`, **checks every SQL string against the schema** and generates the implementation. Observable queries work by **SQLite triggers** that mark a *table* dirty; after each transaction the **InvalidationTracker** re-runs every query that reads that table. The schema version is a contract: every change needs a **migration** or data is lost.

One write, one emission — table-level, after commit



TEMP table, created on Room's connection:

table_id	invalidated
1 Folder	0
3 Note	1

Consequences

- 1000 Inserts in one withTransaction() = **one** re-query; outside it = 1000
- a query JOINing Note + Folder observes **both** tables
- a write to an unrelated Note row still re-queries (table, not row)

Not seen: writes from another process or a second SQLite handle — TEMP triggers live on Room's connection only. Same app, two processes: `enableMultiInstanceInvalidation()`

Codegen & schema

- ksp(room-compiler) generates `AppDatabase_Impl`, `NoteDao_Impl`; a typo'd column is a **build error**. Kotlin codegen is the default on KSP (2.7+); kapt/Java still work in Room 2, **not in Room 3**.
- `@Entity(indices = [Index("folderId")], foreignKeys = [ForeignKey(Folder::class, ["id"], ["folderId"], onDelete = CASCADE)])` — Room warns if an FK column is not indexed (every parent delete would scan). `@Upsert (2.5+)`: insert-or-update.
- `@Embedded` flattens a class into columns (prefix); entities may not hold *references* to other entities — relations are queried, not lazy-loaded.
- `@TypeConverter` pairs (Instant ↔ Long) registered with `@TypeConverters` on the DB, DAO or field.

Example — many-to-many

```

@Entity data class Playlist(@PrimaryKey val id: Long, val name: String)
@Entity data class Song(@PrimaryKey val id: Long, val title: String)
@Entity(primaryKeys = ["playlistId", "songId"], indices = [Index("songId")])
data class PlaylistSong(val playlistId: Long, val songId: Long)
data class PlaylistWithSongs(
    @Embedded val playlist: Playlist,
    @Relation(parentColumn = "id", entityColumn = "id",
        associateBy = Junction(PlaylistSong::class,
            parentColumn = "playlistId", entityColumn = "songId"))
    val songs: List<Song>)
@Dao interface PlaylistDao {
    @Transaction @Query("SELECT * FROM Playlist WHERE id = :id")
    fun observe(id: Long): Flow<PlaylistWithSongs?> // 2 queries
    @Upsert suspend fun upsert(song: Song)
  }

```

`@Relation` = a **second query** per parent batch — without `@Transaction` a write between the two gives a torn read.

Threading & transactions

- A blocking DAO call on the main thread throws `IllegalStateException` (unless `allowMainThreadQueries()`). suspend DAOs hop to Room's executors / query context themselves — no `withContext(IO)` needed.
- Multi-DAO write: `db.withTransaction { ... }` (one writer at a time; readers go on under **WAL**, the default journal mode except on low-RAM devices).

FTS, Paging

`@Fts4(contentEntity = Note::class) @Entity class NoteFts(val title: String) + WHERE NoteFts MATCH :q (join on rowid); Room 3 adds @Fts5. A DAO returning PagingSource<Int, Note> (room-paging) is invalidated by the same tracker; RemoteMediator writes pages into the table.`

Migrations — the version is a contract

- Room stores the schema's **identity hash**; a DB at another version with no path → `IllegalStateException` at open.
- Manual: `val M1_2 = Migration(1, 2) { db -> db.execSQL("ALTER TABLE Song ADD COLUMN len INTEGER NOT NULL DEFAULT 0") }`. Paths chain 1→2→3 or jump 1→3.
- `autoMigrations = [AutoMigration(2, 3, spec = S::class)]` diffs two **exported schema JSONs**; rename vs drop+add needs a spec (`@RenameColumn`, `@RenameTable`, `@DeleteColumn`, `@DeleteTable`). So `exportSchema = true + room { schemaDirectory(...) }`, **JSONs in git**.
- Test: `MigrationTestHelper.createDatabase(1)`, insert rows, `runMigrationsAndValidate(2, M1_2)` = result must equal the v2 schema.
- SQLite ALTER TABLE can't change a type/constraint: new table → copy → drop → rename.

KMP (2.7+) & Room 3

- **2.7.0** (Apr 2025): KMP — Android, iOS, JVM. Builder `.setDriver(BundledSQLiteDriver())` (SQLite compiled from source, same version everywhere) `.setQueryCoroutineContext(Dispatchers.IO); @ConstructedBy + expect object ... : RoomDatabaseConstructor<AppDb>`. Common DAOs: `suspend/Flow` only; `useWriterConnection()`.
- **Room 3.0** (stable 1 Jul 2026, `androidx.room:room3`): KSP + Kotlin only, `SQLiteDriver` only (no `SupportSQLite/Cursor`), `coroutines-first`, `InvalidationTracker.createFlow()`, web targets.

Traps

- ! `fallbackToDestructiveMigration()` in release = a silent **wipe** on any missed path; at most `...OnDowngrade()`.
- ! "Flow emits the changed row" — it re-runs the query, emits *everything*.
- ! New version, schema JSON not committed — no `AutoMigration` later.

Remember: KSP checks SQL · triggers mark tables · commit → re-query · every version needs a tested path.